

Question	Answer	Mark	Mark scheme	Additional guidance
16	39.9	P1	for finding the length of the minor or major arc eg $\frac{220}{360}\pi \times 12$ (= 23(.03834..))	Allow appropriate rounding if calculation seen in parts
		P1	for substituting into the sine or cosine rule to find <i>OD</i> eg $14 \div \sin 140 = OD \div \sin 24$ or ($OD^2 =$) $6^2 + 14^2 - 2 \times 6 \times 14 \times \cos 24$ (=78.5....)	Must involve <i>OD</i> in the relationship but may be implied
		P1	for a complete process to find the length <i>OD</i> eg $14 \div \sin 140 \times \sin 24$ (=8.8(58778..))	
		P1	for a complete process to find the perimeter eg “23(.03834..)” + 14+ “8.8(58778..)” – 6	May be seen in multiple calculations
		A1	for an answer in the range 39.8 to 40	If an answer in the range is seen in working and then incorrectly rounded award full marks.